

MACHINE LEARNING MASTER COURSE

Complete Study Notes: Lessons 1 to 8

Master Project: Universal Customer Churn & Retention Prediction System

1. ML Foundations & Business Problem

Machine Learning discovers rules dynamically from historical data (X , y).

Customer Churn is the universal business problem of customers leaving a service.

2. Questions X vs. Answers y

- **Features (X):** Input information / questions given to the model (Age, Tenure, Bill).
- **Target (y):** Output answer to predict (Churn: 1 = Left, 0 = Stayed).

3. Data Cleaning & Encoding

Cleaning & Encoding Rules

1. **Missing Values (NaN):** Replaced with Median value in `Total_Charges_INR`.
2. **Text Columns:** Converted to 0/1 binary columns using `pd.get_dummies(df, drop_first=True)`.

4. Feature Scaling & Train-Test Split

Numerical columns are scaled using `StandardScaler()` so Mean = 0. Data is then split into **80% Training Set** (Padhai) and **20% Test Set** (Exam).

Complete Master Python Code

```
import pandas as pd
from sklearn.preprocessing import StandardScaler
from sklearn.model_selection import train_test_split

# 1. Load Data
df = pd.read_csv('data/customer_churn_data.csv')

# 2. Impute Missing Values with Median
df['Total_Charges_INR'] = df['Total_Charges_INR'].fillna(df['Total_Charges_INR'].median())

# 3. Categorical Encoding
cat_cols = ['Gender', 'Contract_Type', 'Payment_Method', 'Tech_Support', 'P...']
df_encoded = pd.get_dummies(df, columns=cat_cols, drop_first=True)

# 4. Feature Scaling
scaler = StandardScaler()
num_cols = ['Age', 'Tenure_Months', 'Monthly_Charges_INR', 'Total_Charges_INR']
df_encoded[num_cols] = scaler.fit_transform(df_encoded[num_cols])

# 5. Train-Test Split
X = df_encoded.drop(columns=['CustomerID', 'Churn'])
y = df_encoded['Churn']

X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2, random_state=42)

print("Preprocessed Pipeline Complete!")
print("X_train Shape:", X_train.shape)
print("X_test Shape: ", X_test.shape)
```